From Testimony to Network: Modelling Reported Interactions in the Bologna Inquisition Register (1291–1310)

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Abstract

How did a persecuted religious movement sustain itself socially across north-central Italy in the late thirteenth and early fourteenth centuries? This paper addresses this question through a formal network analysis of the Apostolic movement – a lay religious movement active in North-Central Italy between 1260 and circa 1310 – using the Bologna inquisition register, 1291–1310 (Paolini & Orioli, 1982). Three specific questions guide the analysis: Did ministers occupy structurally distinctive positions within the network? Were preaching and resource exchange interdependent processes of a shared "spiritual–material economy"? And to what extent did kinship and gender shape interaction patterns?
The source is a notarial compilation of interrogations, confessions, and sentences that preserves an unusually detailed record of social interactions – preaching, meeting, hospitality, and exchange – linking itinerant dissident ministers with their local supporters. Like most premodern documentary sources, it does not capture directly observed relations but reported ties: connections narrated by informants, recorded by scribes, and shaped by the institutional logic of inquisitorial investigation.

Using the Computer-Assisted Semantic Text Modelling (CASTEMO) framework (Zbíral et al., 2022, 2026), we transformed testimonies into structured relational data by manually encoding persons, actions, and events as predicate-argument structures. The resulting network comprises 328 nodes and 1,287 edges across multiple relation types – preaching ties, resource flows, kinship, and co-presence. We analysed the network using descriptive measures, degree distribution modelling, a counterfactual node-removal test, and a multilayer Exponential Random Graph Model – ERGM (Lusher et al., 2012).

Across all analyses, the findings point in a consistent direction. Ministers were significantly more likely to initiate preaching ties and acted as bridging hubs between clusters – corroborating and formally testing earlier conclusions by Timberlake-Newell (Timberlake-Newell, 2012) on the structural role of ministers in the same movement. A counterfactual test confirmed that removing the twenty-one ministers disrupts network cohesion far more than removing the twenty-one most-connected supporters. Preaching ties and reverse resource flows strongly co-occurred, supporting the hypothesis of a structural "spiritual–material economy"

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sciencesconf.org:hnr2026:685391

in which spiritual services and material support were reciprocally entangled. Kinship facilitated material support, while gender effects on interaction patterns were minor – men and women displayed moderate and comparable levels of homophily.

Methodologically, the study demonstrates how formal network tools can be applied to premodern sources that document reported rather than observed relations. The findings contribute to longstanding debates on the organisational structure of medieval dissident movements, while the analytical framework is transferable to other inquisitorial registers and premodern trial sources where reported interactions can be systematically encoded. More broadly, the paper argues that moving beyond visualisation and centrality scores toward statistical modelling – including ERGMs – allows historians to test mechanisms rather than merely identify patterns, and to answer historical questions in a more robust and transparent way.

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Keywords: historical network analysis, inquisitorial sources, Bologna, Apostolic movement, Centrality measures, ERGM.

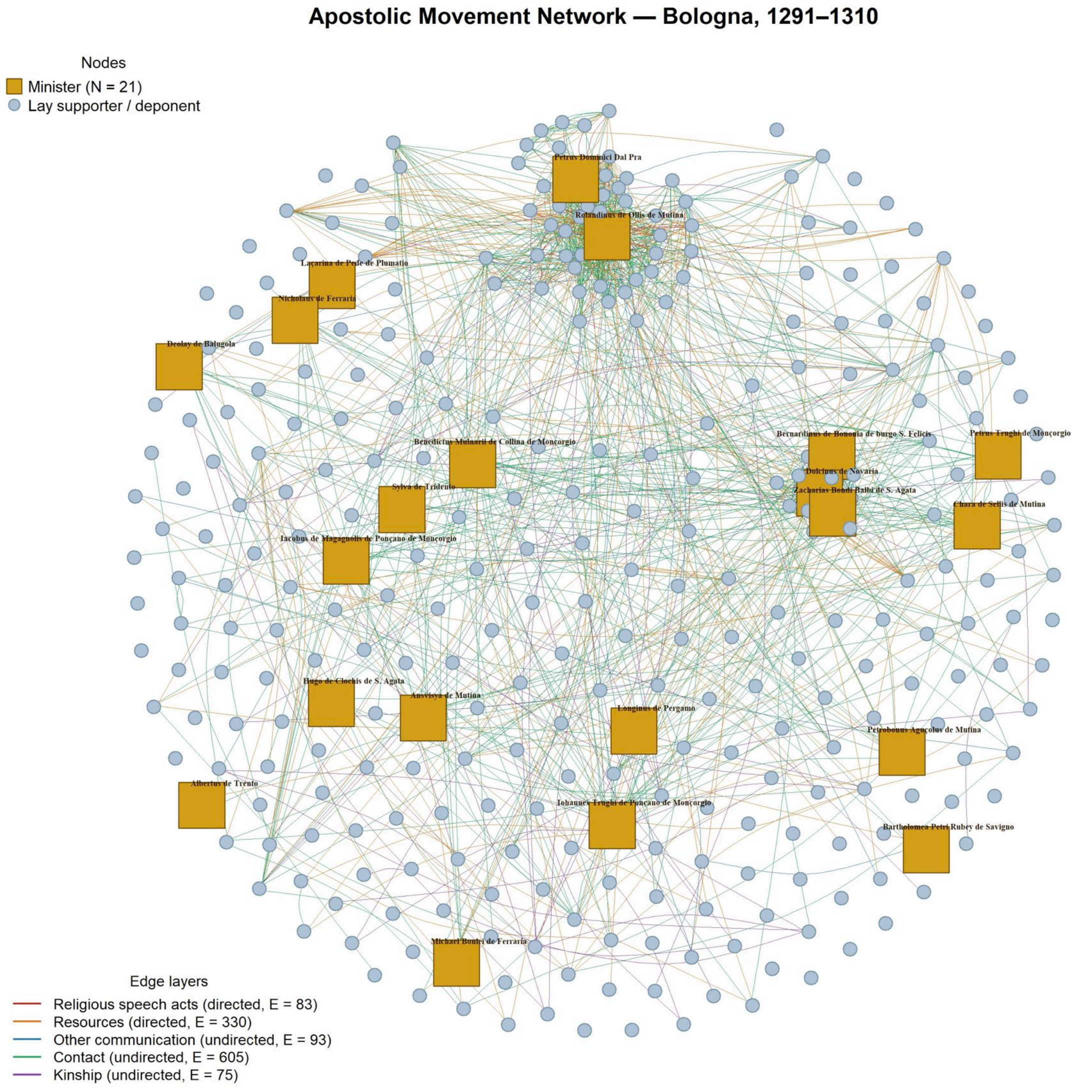

Figure 1. Multilayer network of the Apostolic Movement (Bologna, 1291–1310). All five relation types are overlaid in a single visualisation: religious speech acts (crimson, directed), resource exchange (orange, directed), other communication (blue, undirected), contact (green, undirected), and kinship (purple, undirected). Nodes represent persons documented in the Bologna inquisition register (1291–1310); edges encode ties extracted from testimonies using the CASTEMO framework (Zbíral et al., 2022, 2026). Gold squares mark itinerant dissident ministers (N = 21); grey-blue circles mark lay supporters and deponents. Node labels are shown for ministers only. The layout is a Fruchterman–Reingold arrangement anchored on the religious speech act layer. Edges reported exclusively by deponent-ministers have been removed (sensitivity analysis). Ministers occupy structurally central positions across all relation types, acting as bridges between otherwise disconnected clusters.

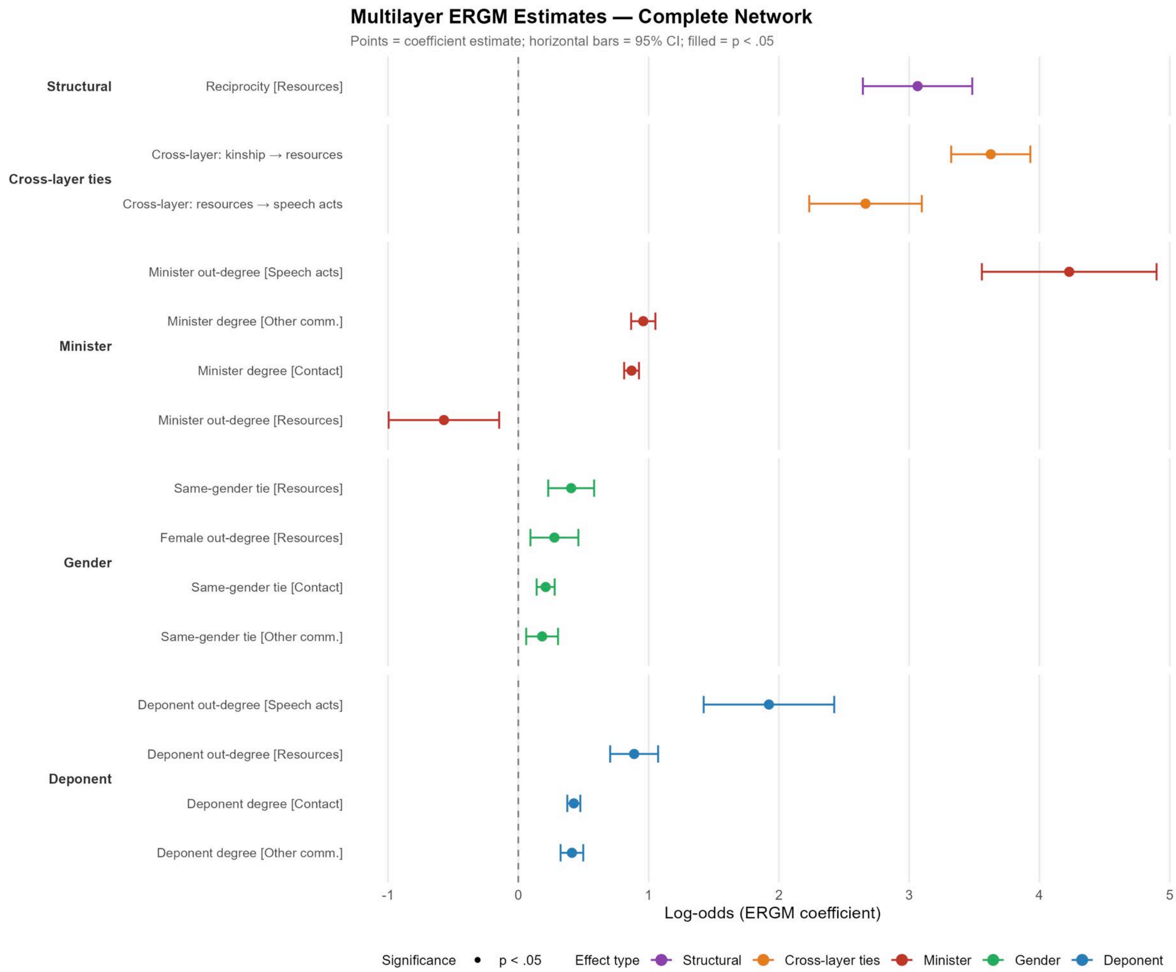

Figure 2. Multilayer ERGM coefficient plot for the complete network. Estimates and 95% confidence intervals from a multilayer Exponential Random Graph Model (ERGM) fitted to the complete five-layer Apostolic Movement network (Bologna inquisition register, 1291–1310). The model includes per-layer density controls (not shown), one structural term (reciprocity in resource exchange), two cross-layer effects, four minister effects, four gender effects, and four deponent effects. Filled symbols indicate terms significant at $p < .05$ (15 of 15 substantive terms). Positive coefficients indicate a higher probability of tie formation; negative coefficients indicate suppression. Ministers show strong positive out-degree effects in religious speech acts and resource exchange, consistent with their role as itinerant preachers who both delivered spiritual services and received material support. The cross-layer coefficient for resources predicting speech acts supports the 'spiritual–material economy' hypothesis: pairs connected by resource flows were substantially more likely to also be connected by preaching ties.

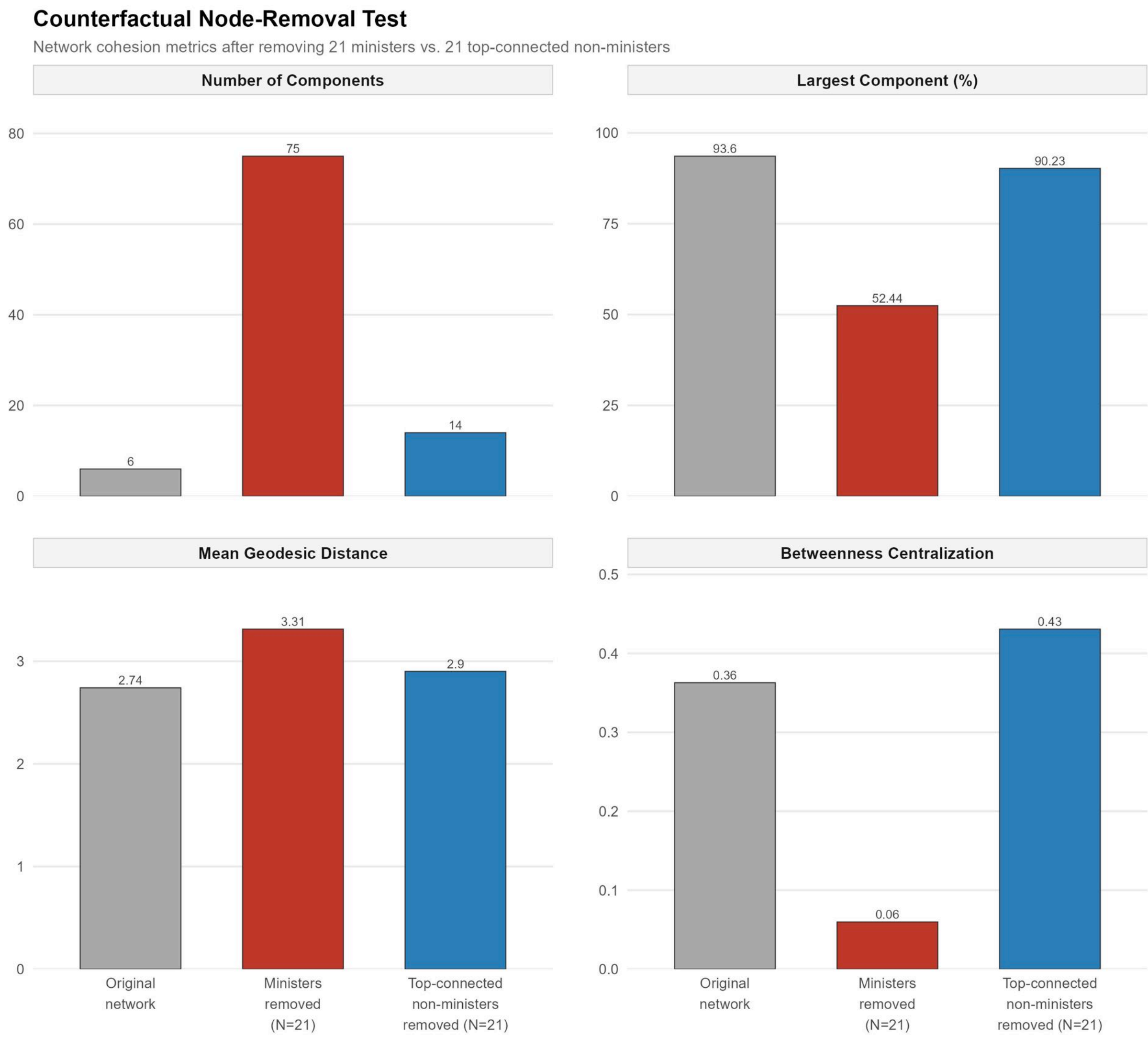

Figure 3. Counterfactual node-removal test: ministers versus top-connected non-ministers. Four network metrics are compared across three scenarios: the original aggregate network (grey), the network after removing all 21 ministers (red), and the network after removing an equivalent number (N = 21) of the most highly connected non-minister nodes (blue). All metrics are computed on the undirected aggregate network combining all five relation types (religious speech acts, resource exchange, other communication, contact, and kinship). Removing ministers produces a substantially larger increase in the number of disconnected components and a greater reduction in the relative size of the largest connected component than removing an equivalent number of the most connected supporters, confirming that ministers functioned as irreplaceable structural brokers whose integrative role exceeded what their raw connectivity alone would predict.